

Multiple Choice

1. **Answer:** C

Options: A) 200.618 g of CaBr₂; B) 175.54 g of CaBr₂; C) 105.32 g of CaBr₂; D) 50.154 g of CaBr₂; E) 150 g of CaBr₂

Note: Correct option: C - 105.32 g of CaBr₂

2. **Answer:** A

Options: A) 15; B) 40; C) 38; D) 13; E) 21

Note: Correct option: A - 15

3. **Answer:** A

Options: A) It causes the pH to fluctuate unpredictably.; B) It causes the pH to equalize with the water's pH.; C) It causes the pH to become extremely basic.; D) It neutralizes the pH completely.; E) It significantly decreases the pH.

Note: Correct option: A - It causes the pH to fluctuate unpredictably.

4. **Answer:** E

Options: A) IR radiation can ionize organic molecules, providing structural information; B) absorption of IR radiation increases with molecular complexity; C) all molecular bonds absorb IR radiation; D) IR spectroscopy can identify isotopes within organic molecules; E) most organic functional groups absorb in a characteristic region of the IR spectrum

Note: Correct option: E - most organic functional groups absorb in a characteristic region of the IR spectrum

5. **Answer:** A

Options: A) 5.9 cm³; B) 12.1 cm³; C) 7.3 cm³; D) 7.9 cm³; E) 9.5 cm³

Note: Correct option: A - 5.9 cm³

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.204 M'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hydrogen fluoride will give a greater yield of product'.

8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'CsF, PH_3'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '-7.44 degrees'.

Long Form Response

11. **Answer:** 1.04×10^{-4} and 6.24×10^{10} l. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 330. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key